Phu Nguyen

My first dataset, Telco Customer Churn, got rejected because it was already clean and there was nothing for me to fix. That taught me the point of this assignment is the cleaning itself, so I switched to the Stroke Prediction dataset, which is much messier.

The biggest thing I learned is that missing data doesn't always look missing. The bmi column had real blanks, but smoking_status used the word "Unknown" for about a third of the patients. I had to turn that into a real missing value before I could fill it.

I also learned why you split the data before doing anything else. I split 70/30 in Python first and only worked on the training part, so the test data stayed untouched.

I learned the difference between encoding, normalization, and scaling. Encoding turns text into numbers, like Male = 0 and Female = 1. Normalization changes the shape of a column, and I used a log transform on glucose to lower its skew from 0.92 to 0.44. Scaling changes the range, and I squeezed age, glucose, and bmi into 0 to 1. When I first opened the final CSV the numbers looked weird, but I figured out they were just scaled values and I could convert them back. I made a second, readable version of the file so people can see real ages and words.

Something I'd do differently is test my choices more. Filling 30% of smoking_status with the most common value is a guess, and capping glucose may have removed real risk signals.

Cyndy Ndong

Working through this assignment completely changed how I look at data preparation because my initial choice, the Telco Customer Churn dataset, got rejected for being already too clean. That early roadblock taught me that the entire point of this project is wrestling with dirty data and fixing it, which led us to switch over to the much messier Stroke Prediction dataset. As we dug into it, I realized that missing data doesn't always look missing at first glance; while the bmi column had real blanks, the smoking_status column hid its missing data by using the word "Unknown" for about a third of the patients, forcing us to convert those into true null values before doing any filling.

I also finally understood why you have to split your data right at the beginning before touching anything else. We set up the 70/30 split in Python first and kept the test data completely untouched while we worked exclusively on the training portion. On top of that, I got a firm grasp on the differences between encoding, normalization, and scaling: encoding translated text into numbers, normalization adjusted distribution shapes like using a log transform on glucose, and scaling squeezed features like age, glucose, and bmi into a 0-to-1 range. When we first checked the final CSV, the numbers looked

bizarre until I realized they were just scaled values, so we made a second readable version of the file. Looking back, if I could change something, I would test my choices more carefully because filling a massive chunk of the smoking status with the most common value is ultimately a guess that might have altered real risk signals.